

SECURITY

Secure Encryption

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The term post-quantum cryptography (PQC) seems to have been coined by Daniel Bernstein around 2006. Interestingly, my interviews on security in this magazine back in 2019/2020 were already peppered with warnings that post-quantum cryptography must be considered and taken into account. Since then, a lot of water has flowed down the river Rhine, and the topic still feels like a barn door for the automotive sector. This makes it all the more interesting to see what insights the articles by IAV, Virtual Vehicle, and, above all, the interview with AISEC shed on the topic this time. The ongoing networking of vehicles is both a blessing and a curse, and compliance with standards such as ISO/SAE 21434 and UNECE R155 etc. is important for mitigating risks, but it is not a panacea when looking beyond the horizon to implement the fastest, most resilient (PQC) cybersecurity possible. Either way, post-quantum cryptography has a crucial role to play: With extremely powerful quantum computers set to hit the market very soon, access to and encryption of data throughout the entire vehicle ecosystem must be quantum-secure. This can also be achieved through hybrid approaches tailored to the current performance capabilities of vehicle computers.

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Cyber Security

Magazine Article: Bartel, F.: Securing the Convergence of OT and IT in the Automotive Sector
www.springerprofessional.de/link/ 50940062

Online Article: Ebert, Ch.; Beck, M.: Generative Artificial Intelligence for Automotive Cybersecurity
www.springerprofessional.de/link/26799546

Magazine Article: Huq, N.: Automotive Cyber Security – Emerging Risks and New Case Study Insights
www.springerprofessional.de/link/27364010

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